

IonQ challenge: Entanglement Distillation

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We propose a heuristic for the IonQ entanglement game. This approach combines a custom parity-check-based distillation circuit with a greedy, easiest-first edge-selection strategy under strict resource constraints. Rather than implementing standard BBPSSW/DEJMPS protocols, we use lightweight error-detection and postselection to enable high-fidelity edge claiming (≥ 0.9). Integrated into a network-level strategy, this method achieves significant connectivity and high performance even with a single Bell-pair budget, demonstrating the effectiveness of adaptive distillation heuristics in constrained quantum networking scenarios. In one representative run, we reached a score of 89 while operating with only one Bell pair of budget for successful claims under the name ‘121234382’.

see Challenge_score_89.ipynb in <https://github.com/Hirmay/iqhack/tree/main>

I. Challenge overview

In this challenge, we design quantum circuits to distill noisy Bell pairs and then use the distilled pairs to claim edges in a network. The goal is to grow a connected subgraph and maximize the total utility while staying within a limited Bell-pair budget.

II. Method description

BBPSSW and DEJMPS are recurrence-based entanglement purification protocols that probabilistically increase the fidelity of shared Bell pairs using only local operations and classical communication (LOCC). Both protocols operate on two noisy entangled pairs at a time, applying bilateral local unitaries and controlled-NOT gates, followed by measurements on one pair and postselection based on correlated outcomes. The BBPSSW protocol is conceptually simpler and is optimized for Werner (isotropic) states, but it is sensitive to asymmetric noise. The DEJMPS protocol generalizes BBPSSW by introducing additional local basis rotations that make it more robust against general Bell-diagonal noise and asymmetric error distributions. In both cases, fidelity is improved at the cost of reduced yield, as unsuccessful measurement outcomes are discarded.

In our algorithm, we implement a custom parity-check-based distillation circuit rather than a standard BBPSSW/DEJMPS protocol. While BBPSSW/DEJMPS use a precisely defined sequence of bilateral CNOTs and full measurements of the sacrificial Bell pair with analytically known acceptance conditions and convergence guarantees, our circuit applies the basis rotations (H and R_x gates) and repeats the same gate block without a formally defined role in the canonical protocols. We measure only one qubit from each side of the ancilla pair and reduces the outcome to a single XOR parity flag, instead of using the full two-bit measurement outcomes as in BBPSSW/DEJMPS. As a result, the algorithm acts as a heuristic error-detection filter whose behavior depends on the input state and post-selection rule, rather than a direct implementation of BBPSSW or DEJMPS with their well-established theoretical guarantees.

III. Strategy for claiming the edges

This routine implements a greedy “easiest-first” edge-claiming strategy under a finite budget (Dijkstra’s inspired strategy). It repeatedly queries the set of currently claimable edges, sorts them by increasing difficulty rating (and then by base-threshold), and selects the single easiest edge to attempt next. The chosen edge is then claimed using an entanglement distillation attempt (attempt-claim-edge-BBPSSW), after which the algorithm refreshes the claimable-edge list; if the list has not changed, it stops early (indicating no progress, e.g., remaining edges are effectively too hard or the state is stagnant). Otherwise it continues, printing success metrics such as achieved fidelity, threshold, and success probability when the claim succeeds, updating the available budget/status each iteration, and terminating when either the budget is depleted or no claimable edges remain.

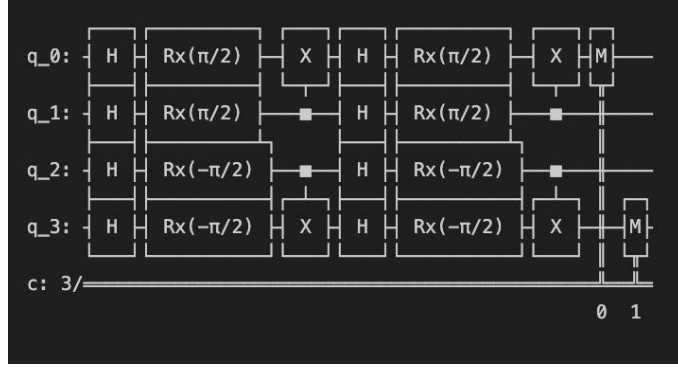


FIG. 1: Quantum scheme of the entanglement distillation.

This routine extends the greedy edge-claiming process by adding a reward-aware target selection step before choosing which edge to attempt. At each iteration it reads the current user status (owned nodes and edges), obtains the set of currently claimable edges, and builds the set of reachable neighbor nodes (nodes incident to a claimable edge connected to an already owned node).

Instead of selecting the next edge purely by difficulty, it ranks candidate reachable nodes using global node attributes, primarily bonus Bell pairs and utility qubits, optionally enforcing minimum requirements for these values and filtering out candidates whose connecting edge exceeds a chosen maximum difficulty. From the remaining reachable candidates, it prioritizes the node with the highest expected gain (highest bonus/reward and utility), and then selects among the edges connecting owned nodes to that preferred target by balancing maximizing the node's bonus/reward while minimizing the edge difficulty (and using base-threshold as a secondary tie-breaker where applicable).

IV. Results

This run represents our strongest overall result, **achieving a score of 89** while operating with the minimal possible resource budget of a single Bell pair, underscoring the robustness of the strategy. Starting from Aachen, Germany, the algorithm expanded the network to 43 owned nodes and 85 edges, with 53 additional edges still claimable, predominantly low-difficulty, high-threshold links (threshold = 0.90, difficulty = 1) across Switzerland, France, and neighboring regions. Beyond raw connectivity, we also reached 9 strategically valuable constrained nodes (including Grenoble, Lyon, Bern, and several UK and Nordic cities), gaining access to bonus Bell pairs and utility qubits that further enhanced network potential. Overall, this performance demonstrates that an easiest-first edge selection combined with high-fidelity entanglement distillation can yield near-optimal expansion, connectivity, and strategic reach even under extremely tight entanglement constraints.

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Player: 121234382 (tutum1)
Score: 89 | Budget: 1 bell pairs
Active: Yes
Starting node: Aachen, Germany
Owned: 43 nodes, 85 edges
Claimable edges: 53
- ['Basel, Switzerland', 'Bern, Switzerland']: threshold=0.90, difficulty=1
- ['Geneva, Switzerland', 'Lyon, France']: threshold=0.90, difficulty=1
- ['Geneva, Switzerland', 'Grenoble, France']: threshold=0.90, difficulty=1
... and 50 more
=====
```

FIG. 2: Table with the final results.

V. Acknowledgement

We'd like to sincerely appreciate the IonQ team, especially Spencer Hill for all his support and insights. We learned a lot from them and were inspired to pursue further.